

BREMERHAVEN, Germany

WMO: 101290

Lat: **53.5333N**Lon: **8.5761E**Elev: **18**StdP: **14.69**Time Zone: **1.00 (EUC)**Period: **82-03**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	15.0	19.4	10.9	9.6	18.0	14.0	11.2	20.8	40.1	46.6	34.3	43.4	8.4	80	0.674

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	81.0	66.0	77.2	64.3	73.6	63.3	68.3	76.9	66.5	73.8	64.9	71.6	9.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.4	93.8	71.6	63.7	88.4	70.2	62.0	83.2	68.6	32.5	77.2	31.1	73.7	29.9	71.8	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.7	27.3	24.4	DB	15.6	88.0	7.5	3.9	10.3	90.8	5.9	93.1	1.7	95.3	-3.7	98.1	(4)
(5)			WB	14.8	72.2	7.3	1.6	9.6	73.3	5.3	74.3	1.3	75.2	-4.0	76.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.1	35.4	35.0	40.8	46.2	54.6	59.6	63.5	63.2	57.9	51.2	42.6	38.4	(6)	
	DBStd	11.85	8.64	7.54	6.13	5.49	6.38	5.83	5.10	4.67	3.91	5.40	5.76	6.18	(7)	
	HDD50	1965	453	421	290	140	23	1	0	0	0	49	227	361	(8)	
	HDD65	5910	918	840	751	564	328	183	91	92	218	428	672	825	(9)	
	CDD50	1644	0	0	4	26	167	289	420	410	236	87	5	0	(10)	
	CDD65	115	0	0	0	0	7	21	46	37	4	0	0	0	(11)	
	CDH74	723	0	0	0	1	70	139	284	211	18	0	0	0	(12)	
	CDH80	151	0	0	0	0	5	33	60	52	1	0	0	0	(13)	
(14) Wind	WSAvg	11.9	14.0	12.4	12.4	11.2	10.9	11.1	11.6	11.1	11.8	11.9	12.1	12.8	(14)	
(15) Precipitation	PrecAvg	33.1	2.6	2.2	2.0	1.6	2.3	3.3	3.7	3.4	3.3	2.9	3.0	3.0	(15)	
	PrecMax	44.3	5.3	4.7	4.2	3.5	5.1	7.4	6.6	7.6	9.6	7.3	5.8	7.8	(16)	
	PrecMin	21.5	0.1	0.9	0.4	0.3	0.6	0.7	1.5	1.3	0.8	1.0	0.3	0.9	(17)	
	PrecStd	5.1	1.4	1.0	1.1	0.7	1.0	1.6	1.6	1.5	2.0	1.5	1.4	1.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.8	55.4	61.2	67.9	80.2	84.6	85.8	85.9	76.5	69.4	56.8	52.8	(19)	
		MCWB	48.5	50.9	52.4	54.2	62.4	67.9	68.9	68.2	65.7	62.0	52.6	49.4	(20)	
	2%	DB	48.7	50.7	54.4	62.8	75.4	77.4	81.0	79.9	70.4	64.6	53.9	50.8	(21)	
		MCWB	46.5	47.0	49.0	52.8	60.5	63.3	67.2	65.9	62.2	58.4	51.0	48.5	(22)	
	5%	DB	47.0	46.9	51.2	58.9	70.9	73.3	76.9	74.7	66.7	61.1	52.2	48.9	(23)	
		MCWB	45.2	43.9	46.9	51.2	58.4	62.8	65.3	64.7	60.5	56.5	50.0	47.1	(24)	
	10%	DB	45.4	44.4	48.7	55.4	65.5	69.5	72.8	71.1	64.3	58.4	50.5	47.0	(25)	
		MCWB	43.6	41.7	45.5	49.7	56.6	61.2	64.3	63.2	59.1	54.7	48.6	45.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.4	51.3	52.8	57.0	64.4	70.0	70.9	69.9	66.7	63.1	53.9	50.3	(27)	
		MCDB	51.2	55.1	59.5	63.8	76.5	80.8	80.0	81.4	73.6	68.5	55.6	51.6	(28)	
	2%	WB	47.0	47.4	49.8	53.8	61.2	66.4	68.7	67.4	63.7	59.3	51.7	48.6	(29)	
		MCDB	48.3	49.9	53.2	60.0	71.9	72.5	77.4	75.3	68.3	63.3	53.4	50.3	(30)	
	5%	WB	45.5	44.4	47.7	52.0	59.2	64.0	66.9	65.9	61.8	57.2	50.2	47.2	(31)	
		MCDB	46.9	46.8	50.3	57.9	68.8	71.0	74.8	72.9	66.2	60.4	51.9	48.7	(32)	
	10%	WB	43.6	42.1	45.9	50.1	57.1	62.0	65.0	64.4	59.9	55.4	48.8	45.5	(33)	
		MCDB	45.2	44.0	48.4	55.2	65.1	68.5	72.0	70.6	63.4	57.7	50.4	46.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	8.0	9.6	11.2	12.4	11.0	11.4	11.6	10.2	9.4	7.4	6.4	(35)	
		MCDBR	7.5	10.6	14.2	18.0	21.7	18.9	19.7	19.3	15.1	13.5	9.4	8.1	(36)	
	5% WB	MCWBR	6.7	8.3	9.5	10.9	11.0	9.8	9.9	9.8	9.6	9.0	8.0	7.5	(37)	
		MCWBR	7.3	10.3	12.6	15.8	19.5	16.3	17.1	15.9	13.8	11.4	9.1	7.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.327	0.346	0.376	0.410	0.405	0.404	0.420	0.413	0.388	0.362	0.340	0.318	(40)	
		taud	2.311	2.318	2.253	2.215	2.271	2.299	2.268	2.297	2.354	2.388	2.343	2.303	(41)	
	Ebn at Noon		200	236	252	259	267	269	261	256	247	227	193	178	(42)	
		Edh at Noon	19	26	34	40	40	39	40	36	30	24	19	16	(43)	
(44) All-Sky Solar Radiation	RadAvg		180	394	770	1274	1593	1664	1590	1317	953	531	226	137	(44)	
	RadStd		20	53	79	153	174	133	211	96	113	57	23	10	(45)	

Historical Trends

		DBAvg	Heating		
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